

CPEN 416

Lecture 07: superdense coding and quantum teleportation

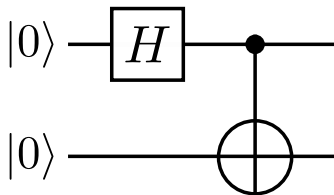
Thursday 1 October 2026

Announcements

- Tutorial assignment 1 due tomorrow 23:59
- Bring me your midterm reference sheets for approval by 5pm tomorrow
- Monday's tutorial: midterm prep

Last time

We introduced the idea of entanglement and entangling gates.



Learning outcomes

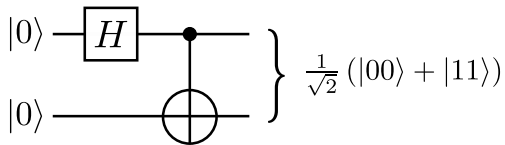
- measure multi-qubit systems in the Bell basis
- leverage entanglement to implement superdense coding
- prove that arbitrary quantum states cannot be “cloned”
- teleport a state from one qubit to another

Bell states

Remember how we created

$$|\psi_{00}\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle),$$

from the $|00\rangle$ state:



Bell states

Exercise: Apply the same circuit to the other 3 computational basis states? What is special about these four states?

Review: basis rotations

Bell basis measurement

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \left\{ \begin{array}{l} \text{---} \bullet \text{---} [H] \text{---} |0\rangle \\ \quad | \oplus \text{---} |0\rangle \end{array} \right.$$

$$\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \left\{ \begin{array}{l} \text{---} \bullet \text{---} [H] \text{---} |0\rangle \\ \quad | \oplus \text{---} |1\rangle \end{array} \right.$$

$$\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \left\{ \begin{array}{l} \text{---} \bullet \text{---} [H] \text{---} |1\rangle \\ \quad | \oplus \text{---} |0\rangle \end{array} \right.$$

$$\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \left\{ \begin{array}{l} \text{---} \bullet \text{---} [H] \text{---} |1\rangle \\ \quad | \oplus \text{---} |1\rangle \end{array} \right.$$

Two famous quantum algorithms, **superdense coding** and **teleportation** work by performing a measurement in the Bell basis.

Superdense coding

Suppose Alice wants to send Bob two classical bits of information, say '1' and '0'.

Q1: How many classical bits must she send to Bob to do this?

Q2: How many *qubits* must she send to Bob to do this?

Superdense coding

Alice and Bob start the protocol with this shared entangled state:

$$|\Phi\rangle_{AB} = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

Next, depending on her bits, Alice performs one of the following operations on her qubit:

$$\begin{aligned} 00 &\rightarrow I \\ 01 &\rightarrow X \\ 10 &\rightarrow Z \\ 11 &\rightarrow ZX \end{aligned}$$

Superdense coding

What happened to the entangled state?

$$|\Phi\rangle_{AB} = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

It will transform to:

00 $\rightarrow I$

01 $\rightarrow X$

10 $\rightarrow Z$

11 $\rightarrow ZX$

Superdense coding

Now, Bob can either

- perform a measurement directly in the Bell basis
- perform a basis transformation from the Bell basis back to the computational basis

to determine his state with certainty, and thus, the bits Alice sent.

$$(H \otimes I) \text{CNOT} \cdot \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) = |00\rangle$$

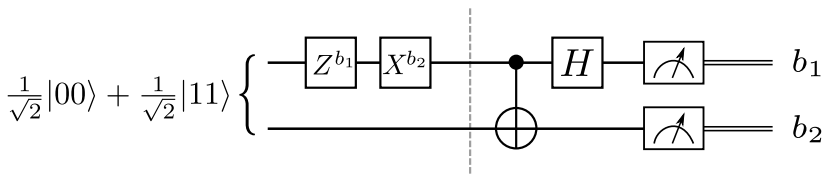
$$(H \otimes I) \text{CNOT} \cdot \frac{1}{\sqrt{2}} (|10\rangle + |01\rangle) = |01\rangle$$

$$(H \otimes I) \text{CNOT} \cdot \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle) = |10\rangle$$

$$(H \otimes I) \text{CNOT} \cdot \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle) = |11\rangle$$

Hands-on: superdense coding

Let's go implement it!



Copying quantum states

Suppose you found a really cool quantum state, and you want to send a copy to a friend. Under what conditions is this possible?

Exercise: develop a circuit that “copies” $|0\rangle$ and $|1\rangle$.

Copying quantum states

Will your circuit work in general? Is there a state for which it fails?

(Not) copying quantum states

The no-cloning theorem

It is impossible to create a copying circuit that works for arbitrary quantum states.

In other words, there is no circuit that sends

for any arbitrary $|\psi\rangle$.

Proof of the no-cloning theorem

Suppose we want to clone a state $|\psi\rangle$. We want a unitary operation that sends

where $|s\rangle$ is some arbitrary state.

Suppose we find one. If our cloning machine is to be universal, we must also be able to clone some other state, $|\varphi\rangle$.

Proof of the no-cloning theorem

We purportedly have:

$$U(|\psi\rangle \otimes |s\rangle) = |\psi\rangle \otimes |\psi\rangle$$

$$U(|\varphi\rangle \otimes |s\rangle) = |\varphi\rangle \otimes |\varphi\rangle$$

Take the inner product of the LHS of both equations:

Now take the inner product of the RHS of both equations:

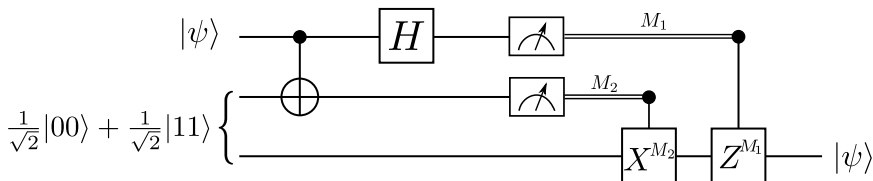
Proof of the no-cloning theorem

For what states does

$$(\langle\psi|\varphi\rangle)^2 = \langle\psi|\varphi\rangle$$

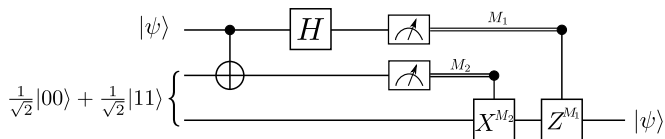
Teleportation

We cannot clone arbitrary qubit states, but we *can* teleport them!



Exercise: determine the state of the system immediately before the measurements.

Teleportation



Quantum teleportation: the details

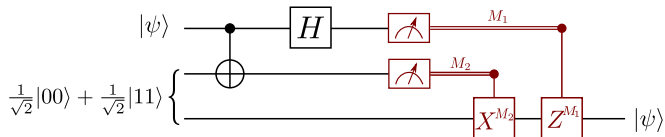
Before measurements, the combined state of the system is

What do you notice about this state?

Quantum teleportation: the details

Alice measures in the computational basis and sends her results to Bob, who can adjust his state as needed.

Let's implement it!



Next time

Content:

- Midterm on 6 October
- After midterm: Oracle-based algorithms (Deutsch and Grover)

Action items:

- 1 Submit tutorial assignment 1
- 2 Study for midterm (practice problems and midterms on PL)
- 3 Submit midterm reference sheet for review by Tuesday 2pm
- 4 Attend Monday's tutorial

Recommended reading (lecture 01-07, and midterm)

- Codebook nodes IQC, SQ, MQ
- Nielsen & Chuang 1.2, 1.3.1-1.3.7, 2.2.3-2.2.5, 2.3, 4.3

Read ahead: Codebook BA, GA; Nielsen & Chuang 1.4, 6.1